



Welcome to CMPT 295

Introduction to Computer Systems

My name is Anne Lavergne

Please, feel free to call me Anne!

Lecture 1 – Course Overview + Activity

Land Acknowledgment

- I would like to acknowledge the fact that this semester, we shall all be learning at SFU Burnaby campus, which is located on the unceded Traditional Coast Salish Lands including
 - the Tsleil-Waututh (sə́lilwə́təʔt),
 - the Kwikwetlem (kwikʷə́ləm),
 - the Squamish (Skwxwú7mesh Úxwumixw) and
 - the Musqueam (xʷməθkʷəy̓əm) Nations.

Today's Menu

- What is CMPT 295?
 - What shall we learn in CMPT 295?
 - What should we already know?
 - Which resources do we have to help us learn all this?
- Activity
- My approach to teaching
- Questions

What is CMPT 295?

- ▶ The SFU Calendar says:
 - ▶ The curriculum [of CMPT 295] introduces students to topics in computer architecture that are considered fundamental to an understanding of the digital systems underpinnings of computer systems. Prerequisite: Either ([MACM 101](#) and ([CMPT 125](#) or [CMPT 135](#))) or ([MATH 151](#) and [CMPT 102](#) for students in an Applied Physics program), all with a minimum grade of C-.

What is CMPT 295?

- The goal of this course is to give us, software developers, a look “under the hood” of a computer, i.e., to learn about Computer Systems, i.e., microprocessor, memory, ...



Download from
Dreamstime.com
This watermark-free image is for personal use only.

16401942
Alexander Kovalevsky | Dreamstime.com

- This knowledge will allow us to become more efficient/effective software developers.

In CMPT 295, we shall learn ...

In a nutshell ... knowing how all of this can impact the execution of our **code** can lead us to write more efficient and reliable **code** and, if necessary, can inform us if/when we optimize our **code**.

C program (.c)

1. Our **code** and **data** are both stored in **memory**. How they are stored in memory and the impact of this storage on the execution of **code**.

Assembly program (.s)

2. As part of learning how our **code** is stored in memory, how a **compiler** transforms our **C code** into **machine code** (i.e., executable) in several steps.

Object (.o) then Executable

3. How a compiler **optimizes** (or not) our **code**.

Computer executes it

CPU

Memory

5. How **memory** is designed and the impact of this design on the execution of our **code**.

4. How a **microprocessor** is designed and how it executes our **code**.

What should we already know?

- Write *correct C programs*
 - C constructs (variables, data types, operations, pointers, conditional statements (if/else), iterative statements (`for` and `while` loops), functions, multi-dimensional arrays, ...)
 - Documentation and testing
- What a **stack** is and how it works
- **Binary/decimal/hexadecimal** numeral systems
 - How to convert from one numeral system to the others
 - Basic arithmetic
- Perform *Boolean algebra* using **and**, **or**, **not**, **xor**

Which resources do we have?

➤ Course web site

<https://www.cs.sfu.ca/CourseCentral/295/alavergn/index.html>

➤ Home page

- Announcements
- Information
- Semester Schedule

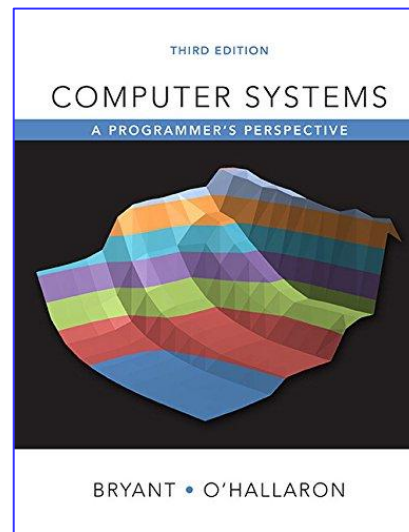
➤ Menu

- Contact Instructor and TAs – Labs and Office Hours
- CourSys
- Discussion Forum
- Course Grading Scheme
- ...

Which resources do we have?

► Textbook

- Computer Systems: A Programmer's Perspective, 3/E, Randal E. Bryant, David R. O'Hallaron, Pearson, 2016



- If you wish to purchase the e-Book format of our textbook from the SFU Bookstore, the **Purchasing instructions** are posted on our course website

Lecture 1 Activity - Goals

Goals:

1. Get to know some students in the class
2. Discover our resources – especially our course web site
3. Warm up your C and your computer knowledge in general

Lecture 1 Activity - Instructions

1. Form teams (max. four (4) people)
2. If you are **enrolled** in this course:
 - Access **Lecture 1 Activity** on **CourSys**

If you are on the **waiting list**:

- Pair up with students that have access to **CourSys**
 - You will not be able to submit your answers to this activity, but do not concern yourself: **this activity is worth 0 marks!**
- 3. This activity ends at 2:10pm (+ 2 minutes of grace time). This means that after **2:12pm**, you will no longer be able to submit your answers.
- 4. Finally, make sure you answer all questions
 - 7 questions to be answered in about 20 minutes. Plan your time accordingly.
- 5. Enjoy!

My approach to teaching

- I believe that ...
 - Learning is **your** responsibility.
 - Giving you plenty of opportunities to learn is **my** responsibility.
 - One very effective and efficient way of learning is to practice what we are learning.
 - If we are not practicing, we are not learning.
 - Therefore, it is my responsibility to give you plenty of opportunities to practice the material we learn in this course (and I will).
 - However, it is up to you to make use of these opportunities.

Question?

- If you are on the *waiting list* ...
 - Hang in there! 😊 Continue attending the lectures.
 - This way, if you get registered into the course, you will not have missed a beat!
 - You will have access to our course web site but not CourSys.
 - If you get registered into the course, it will take a few days before you gain access to CourSys.
 - No grade-based activities due during Week 1 and Week 2.
 - Waiting lists will be resolved by the end of Week 1.
 - If you have not been enrolled into CMPT 125 by the start of Week 2, you may wish to try to enroll next semester.
 - Please, do not send emails to the instructor 😞.
 - If you have any questions about the *waiting list*, please, contact our FAS advisors.
 - Good luck!

Summary

- ✓ What is CMPT 295?
 - ✓ What shall we learn in CMPT 295?
 - ✓ What should we already know?
 - ✓ Which resources do we have to help us learn all this?
- ✓ Activity
- ✓ My approach to teaching
- ✓ Questions

Next Lecture

➤ Data Representation

➤ How to get ready for our next lecture(s):

➤ **Reading:** Read Chapter 1 of textbook (*optional*)

Read Section 2.1 of Chapter 2 (*not so optional* 😊)

➤ As we read sections of our textbook, I strongly encourage you to do the **Practice Problems** in each section (answer are in the back of the chapter)

➤ **Lecture:** Download the partial lecture notes found under the column **Lecture** in the table on our course web site

➤ **Lab:** Do **Lab 0** (to be done before Lecture 2) and **Lab 1** (to be done before Lecture 3) on your own as **review**

